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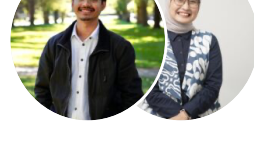
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Burning questions over Indonesia's cooking transition

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IN BRIEF

Indonesia's move to cleaner cooking fuels, achieved largely through the 2007 'Zero Kero Program', has significantly reduced domestic kerosene use but has resulted in a reliance on imported liquified petroleum gas (LPG) and increased the government's LPG subsidy. Now there is an effort to transition to induction stoves, but this has met challenges such as low adoption rates, the need for an expanded renewable energy grid to mitigate fossil fuel dependency, financial and infrastructure implications and cultural norms around cooking methods.



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Indonesia's shift to cleaner cooking fuels has seen notable success, particularly through the 2007 'Zero Kero Program' which reduced domestic kerosene use by 92 per cent in less than ten years by transitioning households' cooking fuel to Liquified Petroleum Gas (LPG). While the program's primary goal was reducing kerosene subsidy costs, it also resulted in positive health impacts through reduced indoor air pollution.

But today, Indonesia faces a new set of challenges. With an annual LPG consumption of approximately eight million tons and a domestic production capacity of only 1.7 million tons, Indonesia's reliance on LPG imports has driven the government's LPG subsidy spending to as high as Rp 80 trillion (US\$4.9 billion) per year. As President Prabowo Subianto's administration looks toward achieving energy self-sufficiency, the current dependence on imported LPG and subsidies' strain on government finances raise pressing questions about Indonesia's path forward.

Indonesia has taken some steps to address the financial burden of LPG subsidies. The Indonesian government aims to reduce imports of LPG to zero by 2030, partly by promoting electric cooking appliances, including induction stoves, in 700,000 households. Yet the adoption rate is currently low — only 0.76 per cent of Indonesian households in 2021 used induction stoves. The challenges of electric stove pilot projects in Denpasar and Semarang in 2022 came from low household electrical capacities which were unsuitable for high-wattage stoves.

The push for induction stoves poses a trade-off — though it may reduce the financial burden of imported LPG, the country still depends heavily on fossil fuels for electricity. Unless there is a massive deployment of renewable energy, this shift could increase greenhouse gas emissions.

For induction stoves to be an environmentally sustainable option, acceleration in the renewable energy transition is the first step. About 14 per cent of the country's energy mix comes from renewable sources, much less than the 23 per cent target set for 2025. Expanding the renewable energy grid would also align with Prabowo's broader goal of achieving energy independence.

With an expanded renewable energy grid, the government could reallocate the LPG subsidies towards incentivising households that are ready to transition to induction stoves, addressing the financial burden of LPG subsidies while simultaneously promoting sustainable energy practices.

Induction stoves can also be incentivised by implementing purchasing restrictions on subsidised LPG to ensure that it only reaches its intended beneficiaries — low-income households. These restrictions, combined with targeted incentives, could accelerate the adoption of induction stoves.

In 2024, the government imposed limitations so that only registered, eligible people could purchase the subsidised 3-kilogram LPG tanks. While the implementation initially appeared inadequate due to subsidised LPG still being accessible to all consumers, the subsidy is largely reaching its intended beneficiaries — most subsidised LPG consumers come from the poorest 70 per cent of households.

Higher demand for induction stoves could also promote domestic production by encouraging new manufacturers to enter the market and enabling existing producers to reduce per unit production costs through larger-scale production. Indonesia's domestic production of induction stoves remains limited with only two manufacturers available and an annual production capacity of 317,000 units in 2022.

To address the low electrical capacities of many Indonesian households, a possible initiative is to promote domestic production of lower-wattage models of induction stoves that are less costly and more compatible with household capacities. But there are costs and barriers associated with low-wattage stoves, including research and development costs and cultural barriers. Since low-wattage could mean slower cooking, Chinese-Indonesian dishes that rely on high-heat wok cooking might not be suitable, while the opposite applies for slow-cooking dishes like the West Sumatran 'rendang'.

Providing incentives in the form of lower costs to upgrade households' electrical capacities can also be implemented. But some lower- to middle-income households might be more reluctant to adopt induction stoves due to higher electricity costs unless they receive continuous incentives such as electricity discounts. Ensuring that households upgrading their electrical capacities also adopt induction stoves — possibly through enforcement mechanisms — should also be factored into policy decisions to maximise the effectiveness of such incentives.

The economic feasibility of switching to induction stoves also depends largely on the payback period — how long it takes households to recover their initial investment through energy cost savings. This investment includes not just the stove itself, but also new cookware compatible with electric cooking. Understanding these costs is crucial for policy design.

Indonesia's transition to induction stoves requires more than policy reforms and infrastructure improvements. It demands a fundamental transformation of the country's energy mix. With renewable energy making up only 14 per cent of the current mix, sustainable adoption appears unlikely in the near future. Success hinges on a coordinated approach — expanding the renewable energy grid first, then gradually implementing household transitions and addressing infrastructure gaps while protecting vulnerable households. Only then can Indonesia achieve energy independence without compromising environmental objectives or household welfare.

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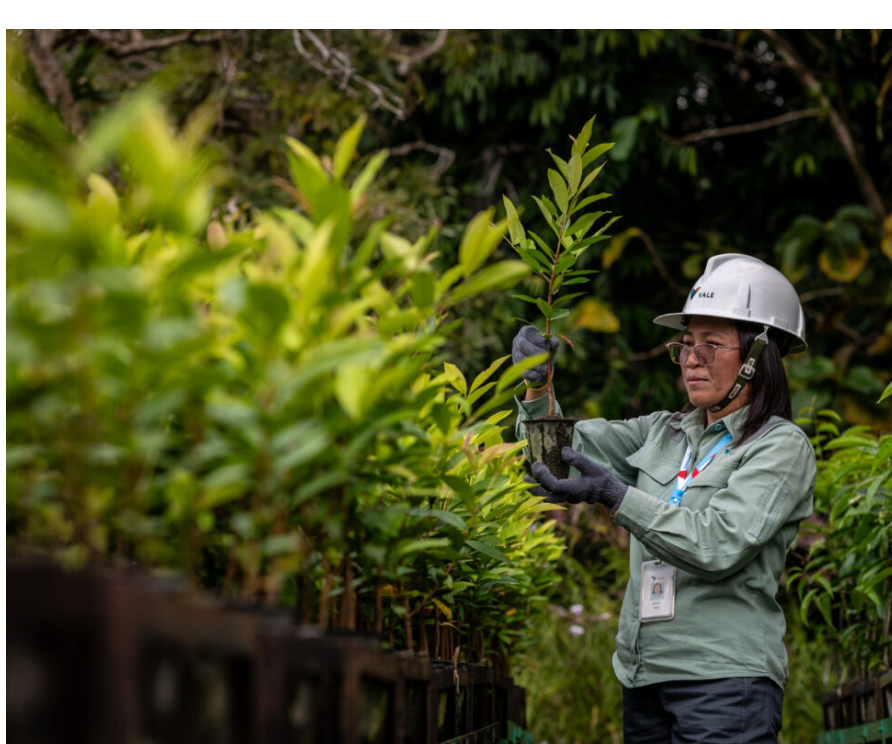
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